

MEDICAL DOMAIN

CARD IDEA:

so in pakistan we have prescriptions. like manual prescriptions. unlike abroad we dont have a central system. i want a card based system like cnic that every person must have a health card. that health card is basically unique stating the users id number. (like cnic number). there is a qr code on that card. and like that qr code have all the history of the user from head to toe. all prescribed medications previous history current medications doctors that have been consulted for each medication that will also be easy to yk connect the dots if like a person have history of some symptops of a disease then we can infer it. so basically i want a card based system. i want mobile and ios app that does this. is centralised supports english urdu. have different logins for patient and doctor and admin. and to log in you need to put in your cnic with password. no email thing. to create an account you need phone number cnic and password and otp thing. i want you to research the market. the medical domain in pakistan. find resources think as a cofounder. tell me the if it is a good idea or not. i want you to be honest.

Feasibility of a CNIC-based Digital Health Card System in Pakistan (2026)

Introduction

Pakistan's health sector remains fragmented; patients receive separate files at each hospital and there is no central record, making continuity of care difficult. Out-of-pocket spending accounts for **56 %** of health costs, and public health expenditure is low. Telemedicine platforms and e-pharmacies (e.g., Sehat Kahani, Healthwire, Dawaai,

Oladoc, Marham, MediQ and InstaCare) have emerged, but they operate independently and do not share records. You propose a **CNIC-linked health card** with a QR code containing each person's entire medical history. The system would include patient, doctor and admin log-ins, work in Urdu and English, and use CNIC and OTP rather than email. This report assesses the idea's viability by examining market conditions, digital inclusion, existing initiatives, regulation, opportunities and risks.

Market and Digital Inclusion

Digital & Mobile Access

- **Smartphone penetration** – According to the **OICCI Digital Report 2025**, only **73 million people** (≈31 % of Pakistan's population) used smartphones and the rest relied on basic phones; the report notes that smartphone ownership is especially low among women (48 % gender gap).
- **2026 Economic Survey** – Government data released in mid-2026 showed that **smartphone adoption reached 71.6 %**, yet **28 % of mobile users remain on 2**



G networks phoneworld.com.pk. While the figure suggests rapid growth, connectivity remains uneven.

- **Internet access** – The federal **Household Integrated Economic Survey (HIES) 2026** reported that **70 % of households had internet access and 57 % of individuals used the internet**. However, the digital divide persists; power outages, diesel theft at mobile towers and 2 G reliance limit reliable connectivity



phoneworld.com.pk.

- **Mobile connections** – GSMA Intelligence recorded **194 million mobile connections** in late 2025, equal to 75.9 % of the population, but many people use multiple SIMs.
- **Digital literacy** – A **Gallup Pakistan survey** (2020) found that **55 % of Pakistanis did not own a personal mobile/smartphone and only 20 % had**



ever used one, with female non-ownership at 74 % gallup.com.pk.

Digital skills and literacy remain obstacles, particularly for rural women and the elderly.

Implications: The proposed health card depends on smartphone-based apps and internet connectivity. While adoption is improving, a large share of Pakistan's population still lacks smartphones or consistent access. Solutions must include offline or USSD options and invest in digital literacy and infrastructure.

Health-tech and Telemedicine Market

The past five years have seen rapid growth in Pakistan's health-tech sector:

Platform (year launched)	Core services	Scale / funding
Sehat Kahani (2017)	Telemedicine app connecting patients to female doctors; on-demand lab tests and medicine delivery.	Raised US\$2.7 M Series A in Dec 2023. Reported 141 % YoY growth, 2.6 million consultations, 7,500+ healthcare professionals and 62 e-health clinics .
MedIQ (2020)	B2B/B2B2C hybrid platform integrating telehealth, e-pharmacy and AI-powered facility digitisation.	Raised US\$6 M Series A in 2025 to expand to Saudi Arabia and Qatar; aims to become the healthcare "operating system" for MENAP.
Healthwire (2015)	Book in-clinic or online appointments; order medicines; schedule lab tests; connect with 10,000+ doctors ; partnerships with labs and hospitals.	
Dawaai (2014)	e-pharmacy with online consultations, home lab testing,	

prescription verification; partners with major pharmaceutical brands.

- Oladoc (2015)** Book video/in-person consultations; lab tests; surgery bookings; share medical records; claims **50 million patients** and **25,000 doctors**.
- InstaCare / SuperHealth App (2017)** Provides EMR/EHR, clinic and hospital management systems, telemedicine and pharmacy management software. Claims **4,300 deployments across 50+ cities**, serving **3,000+ healthcare facilities** and **0.5 million patients**.
- Marham (2016)** Mobile app to find doctors by specialty, area, ratings, gender and fee; offers video consultations, a “Call Doctor Now” hotline and a forum for health queries.
- MedznMore (2020)** Pharmaceutical delivery platform; raised **US\$11.5 M pre-Series A**; aims to eradicate counterfeit medicines by working directly with suppliers and operating cold-chain warehouses delivering **100,000+ products per day**.
- Sehat Sahulat Program (SSP)** Government health insurance providing free in-patient treatment up to **PKR 720,000 per family per**

year; by early 2020 covered 6.5 million families.

There are more than **450 health-tech platforms** across categories (teleconsultation, e-pharmacy, appointment booking, EMR/EHR). The market is competitive and dynamic; major players already provide online medical records, appointment booking and telemedicine features similar to those proposed for the health-card system.

Policy and Regulatory Environment

National Digital Health Framework

Pakistan's **National Digital Health Framework 2022–2030** seeks to develop a national health information infrastructure, but implementation is fragmented. Public health facilities often lack interoperable information systems and trained digital health professionals. Private and NGO projects operate autonomously; adoption is uneven across provinces. Stakeholder alignment, investment, digital literacy and robust governance are identified as prerequisites for success.

One Patient One ID

In 2025 the Ministry of Health and NADRA launched the **One Patient One ID** initiative. Each citizen's **CNIC** number serves as their **permanent medical record (MR) number**, enabling consistent access to a unified patient history across hospitals. The project aims to integrate fragmented records, support telemedicine, and improve surveillance and policymaking. Challenges include ensuring data privacy, adoption in rural areas and training of healthcare personnel. The initiative signals government willingness to centralise medical records, making it an important ally for any private health-card project.

Telemedicine Regulation

Pakistan lacks a national telemedicine law, but the **Sindh Telemedicine & Telehealth Act 2021** provides a template. Key provisions include:

- Only **registered medical practitioners** and **health-care service organisations**



may practice telemedicine after appropriate training and registration sindhlaws.gov.pk. A registry must be maintained.

- **Consent and privacy** – Patients must provide explicit consent; providers must inform patients of their qualifications and ensure that platforms used are secure. Data may not be shared without patient consent; breaches of confidentiality may



lead to penalties and imprisonment sindhlaws.gov.pk. Providers must exercise professional judgment to determine whether telemedicine or in-person



care is appropriate sindhlaws.gov.pk.

- **Patient data and records** – Doctors must obtain adequate history and maintain secure electronic medical records. Physical examination is required when



necessary; e-prescriptions must be provided via secure systems



sindhlaws.gov.pk sindhlaws.gov.pk.

- **Prescription restrictions** – Medicines with potential for abuse (e.g., narcotics)



cannot be prescribed remotely sindhlaws.gov.pk.

While only binding in Sindh, the Act demonstrates that telemedicine is being regulated; similar legislation is likely to follow nationally. Any health-card system must align with these rules.

Data Protection and Privacy

Pakistan's **Personal Data Protection Bill 2023** (PDPB) introduces strict obligations for organisations processing personal and **sensitive data** (including CNIC numbers and health information). Key features:

- Applies to any organisation established or operating in Pakistan or offering services to persons in Pakistan. Sensitive personal data includes health data, CNIC and biometric identifiers.
- Establishes a **National Commission for Personal Data Protection**; data controllers must register and may be required to appoint a **Data Protection Officer**. Breaches must be reported within **72 hours**.
- **Data residency** – Critical personal data must be stored on servers located in Pakistan; cross-border transfer is permitted only to countries with adequate legal regimes or under binding contracts.
- Grants individuals the **right to erasure, access and data portability**. Violations can incur heavy penalties.

Any system processing health records must comply with the PDPB: obtain informed consent, ensure data security, host servers within Pakistan, provide deletion and access rights, and appoint a data protection officer. Integrating with NADRA also requires rigorous security since CNIC data are sensitive.

Global Context: Rwanda's One Patient One Record

Rwanda is often cited as a model for integrated digital health. Its **One Patient One Record** programme consolidates patient information across facilities by giving each citizen a **unique patient identifier** and connecting providers via the **Rwanda Health Information Exchange**. Healthcare providers can access a patient's history in real time through the national exchange, improving continuity of care and reducing duplicate tests. The initiative is supported by robust digital infrastructure, governance and partnerships. Rwanda's experience shows that centralised health records are feasible when government leadership, interoperable standards, data privacy and human-centred design are prioritised. It also highlights the importance of training healthcare workers and ensuring rural connectivity.

Opportunities for a CNIC-based Health Card in Pakistan

1. **Alignment with Government Policy:** The government's **One Patient One ID** indicates a desire to unify health records. A private system could provide the technical platform and user-facing apps, complementing government efforts.
2. **Improved Continuity of Care:** A QR-coded card linking to a central record would allow doctors to see a patient's full history, reducing misdiagnoses, duplicate tests and drug interactions. This addresses the current fragmentation where patients start afresh at each facility.

3. **Clinical Decision Support & AI:** Aggregated data could enable analytics to detect disease patterns and flag risky drug interactions, enhancing quality of care. Sehat Kahani plans to develop decision-support tools; a health-card platform could incorporate similar functionality.
4. **Telemedicine & Remote Care:** The card and mobile app could integrate teleconsultation, e-pharmacy and lab services, replicating features of existing platforms while providing a unified record. This could support continuity for patients who migrate between rural and urban areas.
5. **Potential Integration with Insurance:** Linking the card to government insurance programmes (e.g., **Sehat Sahulat Program**) could streamline claims processing and ensure that benefits follow the patient rather than the hospital.
6. **Public Health & Research:** A centralised database could help authorities monitor disease outbreaks, evaluate interventions and plan resource allocation, similar to Rwanda's health information exchange.

Challenges and Risks

1. **Digital Divide & Accessibility:** A large portion of Pakistan's population either



does not own smartphones or relies on 2 G networks phoneworld.com.pk. Requiring users to download an app and scan QR codes may exclude rural residents, women and low-income individuals. Solutions must include offline modalities (e.g., smartcards readable by clinics) and support for USSD or SMS.

2. **Infrastructure Reliability:** Frequent power outages, diesel theft at telecom



towers and limited broadband coverage hinder consistent connectivity phoneworld.com.pk. Without reliable networks, digital records may be inaccessible when needed.

3. **Competition:** Established players like Healthwire, Oladoc, Sehat Kahani, MediQ and InstaCare already offer teleconsultation, appointment booking and e-pharmacy services to millions of users. Many also provide digital medical records on their platforms. Capturing market share will require significant differentiation or government endorsement.
4. **Regulatory Compliance:** Handling CNICs and medical data triggers strict obligations under the PDPB and the Sindh Telemedicine Act. The system must ensure explicit consent, data minimisation, secure servers in Pakistan, audit trails

and the ability to purge data when requested. Non-compliance could lead to fines or criminal liability.

5. **Integration with Healthcare Providers:** To be useful, the card must be accepted by hospitals, clinics and pharmacies. Many public hospitals use paper records; private providers have their own systems. Creating interfaces and persuading providers to adopt a new platform would require resources, incentives and training. Without government mandates, adoption may be slow.
6. **Costs & Sustainability:** Developing secure apps, an interoperable database, and issuing physical cards would require substantial capital. Monetisation could come from subscription fees, transaction fees or insurance partnerships; however, low willingness to pay may limit revenue. Grants or public-private partnerships might be necessary.
7. **Trust & Awareness:** Public trust in sharing health data is low. Education campaigns and transparent governance will be needed to convince patients that their data will remain confidential and will not be misused.

Assessment and Recommendations

Is it a good idea? The concept of a CNIC-linked health card aligns with Pakistan's policy direction and global best practices. Centralised medical records could substantially improve quality of care, reduce costs and support public health. However, the idea's success depends on **government partnership**, **robust infrastructure** and **inclusive design**. Creating such a system as a purely private venture would face serious challenges due to regulatory compliance, competition and adoption barriers.

Recommendations for Proceeding

1. **Partner with Government and NADRA:**
 - Engage with the Ministry of Health and NADRA to align with the **One Patient One ID** program. Government backing would provide legitimacy, access to CNIC databases, and the ability to mandate adoption by public hospitals. Public-private partnership could mirror Rwanda's approach.
 - Integrate with the **Sehat Sahulat Program** to link insurance benefits to the health card.
2. **Build for Inclusivity:**
 - Offer **offline smartcards** with QR codes that can be scanned by clinics using basic devices. Provide **USSD/SMS interfaces** for appointment booking and prescription refills so that users on 2 G networks can participate.

- Localise the user interface in **Urdu and regional languages** and provide voice-based assistance for low-literacy users.
- 3. **Adhere to Data Protection & Telemedicine Laws:**
 - Conduct a **privacy impact assessment** and appoint a **Data Protection Officer** as required by the PDPB.
 - Ensure that servers are located in Pakistan, implement encryption, two-factor authentication, audit logs and role-based access.
 - Obtain explicit patient consent for data sharing and provide mechanisms for data access and deletion.
- 4. **Integrate with Existing Health-tech Platforms:**
 - Rather than competing head-on, explore **API partnerships** with platforms like Oladoc, Healthwire, Sehat Kahani, MedIQ and InstaCare. Allow patients to port their records into the central card, avoiding duplication.
 - Provide value-added services such as **drug–drug interaction alerts**, **chronic disease management** and **AI-based triage**, differentiating your platform from appointment-only competitors.
- 5. **Pilot and Iterate:**
 - Start with a **pilot in one province** or a network of private hospitals to validate workflows. Use feedback from doctors and patients to refine the user experience and address technical or regulatory issues.
 - Measure success through metrics such as reduced duplicate tests, improved adherence, patient satisfaction and physician uptake.
- 6. **Secure Funding and Build Capacity:**
 - Pursue investment from impact investors, local venture funds or international donors interested in digital health and public health.
 - Invest in **training healthcare providers** and **digital literacy programs** to ensure effective use of the system.

Conclusion

A CNIC-based digital health card with QR-coded records could transform Pakistan’s healthcare system by enabling continuity of care, telemedicine integration and evidence-based policymaking. The idea resonates with global initiatives such as Rwanda’s **One Patient One Record** and is consistent with the government’s **One Patient One ID** program. However, success hinges on bridging the digital divide, complying with data protection laws, collaborating with existing health-tech players and securing government partnership. As a co-founder, you should view this as a **long-term, public–private venture** rather than a typical startup; proceed cautiously by engaging stakeholders, piloting incrementally and designing for inclusivity.

Proposal for CNIC-based Health Card App: Front-End & Back-End Design

1 Project overview

Pakistan's health-care system is fragmented and relies heavily on paper-based prescribing. The government is already piloting a **One Patient One ID** program that uses the **CNIC number** as a permanent medical record number. Building a centralised, card-based health information system that works on smartphones and web browsers could improve continuity of care, support telemedicine and reduce duplication. This proposal details the functional and technical requirements for such a system – including separate apps for patients and doctors, an admin web dashboard, CNIC-based authentication, a QR-coded health card, and real-time integration with electronic health record (EHR) systems using the HL7 FHIR standard.

2 Objectives

1. **Universal health record** – create a secure, longitudinal medical history linked to the patient's CNIC that can be accessed by authorised doctors anywhere in Pakistan.
2. **Ease of use** – design patient and doctor apps for Android and iOS with localised UI (English and Urdu) and offline caching to accommodate users without constant connectivity.
3. **Real-time update and viewing** – allow doctors to view a patient's history and add diagnoses, symptoms, medications and dosages that immediately appear on the patient side.
4. **Compliance** – adhere to Pakistan's emerging **Personal Data Protection Bill** (PDPB 2023) which regulates handling of sensitive data (including CNIC numbers and health information) and requires breach notification and data residency. Follow international best practices (HIPAA/HITECH) for encryption, audit logging and authentication.

3 Regulatory & ethical considerations

- **Sensitive data classification:** Under the PDPB, health data, CNIC numbers and biometrics are classified as *sensitive personal data*. The platform must collect the minimum necessary data, obtain explicit consent and allow patients to request deletion (right to erasure within 14 days).

- **Data residency:** Critical personal data must be processed on servers located in Pakistan. Cloud hosting should therefore use local data centres or a hybrid on-premises/cloud setup.
- **Security & risk management:** Conduct a risk assessment and implement technical safeguards such as encryption, multi-factor authentication (MFA), role-based access control (RBAC) and audit logging. A technical blueprint for secure patient portals recommends MFA with passwords and OTPs/biometrics, RBAC to restrict data by user type, session timeouts, AES-256 encryption at rest



and TLS in transit, and comprehensive audit logs disolutions.net.

- **Consent management:** Use a standard consent form for data sharing and ensure versioned consent is recorded, as recommended in telemedicine system design.

4 Functional requirements

4.1 User roles

Role	Capabilities
Patient	Register using CNIC, phone and password; receive OTP for verification; view personal demographic data; scan their health card QR code to add themselves; view chronological health timeline (diagnoses, symptoms, medications, lab results); book appointments and receive reminders; securely message doctors; request prescription refills; update contact information; choose privacy settings (e.g., hide sensitive visits).
Doctor	Register with PMDC number and CNIC after verification by the admin; search or scan patient health cards; view patient summaries; add or edit diagnoses, symptoms, medications and dosages (FHIR Observation , Condition and MedicationRequest resources); upload lab results or attach documents; add notes; communicate with

patients via secure messaging; schedule follow-up appointments; access analytics on their panel.

Admin (Health authority)	Manage doctor credentialing; approve or deny account requests; assign roles; manage health card issuance and QR generation; configure EHR integrations; run reports; manage consent policies and system settings.
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4.2 Core user journeys

1. Account creation & login

- Patient provides CNIC, phone number and password; the system sends an OTP via SMS to verify identity; once verified, a QR-coded digital health card is generated and optionally printed.
- Doctors provide PMDC number, CNIC and credentials; admin reviews and approves; doctors then set a password and complete MFA setup.
- On login, users enter CNIC and password; a one-time code is sent; user must complete MFA for high-risk actions (e.g., modifying data).

2. Viewing health history

- Patient opens the app and sees a chronological timeline of visits, diagnoses and medications. Each entry links to FHIR resources.
- Doctor scans the patient's QR code or searches by CNIC; the app requests the FHIR server for the patient's **Patient** resource and associated **Encounter**, **Condition**, **Observation** and **MedicationStatement** resources; the doctor views the record and adds notes.

3. Entering clinical data

- After a consultation, the doctor selects the patient and uses forms to record symptoms, diagnosis, treatment plans and medication dosages. Each form maps to FHIR resources and is validated. The update is sent via API to the FHIR server; through FHIR **subscriptions** the patient app receives a notification and updates its local copy.

4. Messaging & follow-ups

- Secure messaging is available to ask non-urgent questions, refill prescriptions or provide lifestyle advice. According to healthit.gov, secure messaging improves care coordination and allows patients to communicate at their convenience.

- Patients can schedule appointments and receive reminders; doctors can manage their appointment calendar.

4.3 Additional modules

- **Appointment scheduling & reminders:** As noted in patient portal best practices, portals should allow patients to schedule, change or cancel appointments and receive reminders via SMS or push notifications.
- **Medication refill & adherence:** Patients can request refills and receive alerts when it's time to renew medication.
- **Insurance & billing integration:** While not required initially, support for displaying insurance coverage and claims can be added later.
- **Telehealth integration:** Provide video consultations using secure WebRTC; integrate with existing telemedicine platforms (e.g., Sehat Kahani) via APIs. Ensure that notifications do not reveal sensitive details (e.g., do not mention psychiatry visits).
- **Reporting & analytics:** Admins and doctors can run reports on patient demographics, disease trends and medication adherence for decision support.

5 Technical architecture

5.1 Overall architecture

Use a modular, **FHIR-first** architecture so that all clinical data are stored and exchanged as FHIR resources. A FHIR-first design builds the application on FHIR API and data models from the start, enabling structured, modular data and easy integration with other systems. Key components include:

1. **Mobile apps (Android/iOS)**
 - **Patient App:** Cross-platform (Flutter or React Native) for iOS and Android. UI supports English/Urdu and right-to-left text. Offline caching of the patient's record for areas with poor connectivity; when offline, updates are queued and synced once connected.
 - **Doctor App:** Designed for physicians; supports complex workflows (e.g., capturing detailed clinical data, prescribing) and quick scanning of QR codes. Design separate patient and provider apps because their UI requirements differ dramatically.
2. **Admin Web Portal** – built using a modern JS framework (React or Angular). Provides dashboards for managing users, generating health cards, monitoring usage and configuring integrations.

3. **API Gateway** – central entry point for all client requests. Provides rate limiting, routing, authentication and load balancing.
4. **Identity and Consent Service** – handles CNIC-based authentication, MFA (password + OTP/biometric), session management, RBAC and consent management. Based on OpenID Connect/SMART on FHIR for standardised authentication and authorisation.
5. **FHIR Server** – hosts all medical data as FHIR resources. Use an open-source server like **HAPI FHIR**, which is widely used, interoperable and can be deployed via Docker. HAPI FHIR uses RESTful APIs and supports both JSON and XML. Deploy at least two instances for redundancy and host them on local data centres to comply with data residency requirements.
6. **Application Services** – microservices built with Node.js/Express (or Spring Boot) that implement domain logic: patient management, doctor management, appointment scheduling, messaging, notifications, and reporting. Node.js provides a unified development environment and is lightweight; Express simplifies routing and HTTP operations.
7. **Integration Layer** – connectors for:
 - **NADRA CNIC verification**: call NADRA API to verify CNIC and fetch demographic data.
 - **Laboratory Information Systems (LIS), Radiology (DICOM) and Pharmacy**: integrate via FHIR **Observation**, **DiagnosticReport** and **MedicationDispense** resources.
 - **Telemedicine platforms**: integrate with existing telehealth providers (e.g., Sehat Kahani) to allow video calls.
8. **Messaging & Notification Service** – provides secure message threads between doctors and patients, and sends push/SMS notifications. It logs all interactions in the FHIR server (e.g., **Communication** resource) and ensures messages are encrypted end-to-end.
9. **Analytics & Audit Service** – collects system logs, monitors access patterns and generates audit trails. It provides dashboards for administrators and ensures all access to patient data is logged for compliance (who viewed/modified which resource and when).

5.2 Data model

- Each **Patient** is identified by a unique CNIC; the FHIR **Patient.identifier** uses `system="https://nadra.gov.pk/cnic"` with the CNIC number. Additional identifiers (e.g., insurance numbers) can be added later.
- **Encounters** store visits; **Encounter.subject** references the patient and **Encounter.participant** references the doctor.

- **Conditions** describe diagnoses; **Condition.code** uses SNOMED CT or ICD-10 codes for standardisation.
- **Observations** capture vitals, lab results and symptoms; **Observation.valueQuantity** records measurements.
- **MedicationRequest** stores prescribed medicines and dosages; **Medication** resources describe the drug (brand and generic names). When a doctor updates medications, a new **MedicationRequest** is created and the patient app is notified via FHIR subscriptions.
- **Communication** resources represent secure messages.
- **Provenance** resources track who created/updated each resource and when, providing an audit trail.

5.3 Security & compliance

1. **Authentication & MFA** – Combine password authentication with **one-time passwords (OTPs)** via SMS; optionally add biometric authentication (e.g.,



fingerprint or facial recognition) on devices that support it
disolutions.net.

2. **Role-Based Access Control (RBAC)** – assign roles (patient, doctor, admin). Doctors can only access records of patients they are treating. Patients can only view their own record. Admins cannot view clinical details but can manage metadata. The DI Solutions blueprint emphasises RBAC to restrict data visibility



based on user type
disolutions.net.

3. **Encryption** – Use **TLS/HTTPS** for all data in transit and **AES-256** encryption for



data at rest
disolutions.net. Protect cryptographic keys in a hardware security module (HSM).

4. **Audit logging & monitoring** – Log user logins, failed attempts, data



accessed/modified and new device access
disolutions.net. Provide monitoring dashboards for suspicious activities.

5. **Session management** – Implement session timeouts and automatic logout after



periods of inactivity disolutions.net.

6. **Risk assessment & compliance** – Conduct regular security audits and penetration tests; evaluate vulnerabilities; keep software dependencies updated. Use frameworks that support HIPAA and PDPB compliance. DI Solutions recommends risk assessments, choosing compliant technology stacks (e.g., PostgreSQL with built-in encryption), implementing technical safeguards,



designing accessible UX and partnering with trusted vendors disolutions.net.

7. **Consent & privacy** – Provide transparency on data use; log patient consents; allow patients to revoke access. Store only the minimum necessary data; never reveal details in push notifications (e.g., appointment type).

6 Front-end design considerations

1. **Cross-platform development** – Use **Flutter** or **React Native** so that one codebase serves Android and iOS, reducing development cost. Use **Localization** packages to support English and Urdu; allow users to switch language at runtime.
2. **Accessibility** – Follow WCAG 2.1 Level AA guidelines for contrast, font size and voice-over support. For low-literacy users, provide icons and audio prompts. For left-to-right (Urdu) support, ensure UI can flip direction.
3. **Offline support & data sync** – Implement a local database (e.g., SQLite or Realm) to store a read-only snapshot of the patient's record. Use FHIR **Subscription** mechanisms to synchronise data when connectivity returns.
4. **QR scanning & generation** – Integrate camera access to scan patient health cards. The QR code encodes the CNIC and possibly an opaque token; scanning triggers fetching the patient's record via the API. Use libraries like [react-native-camera](#) or [Flutter Barcode Scanner](#).
5. **Forms & validation** – Create dynamic forms mapped to FHIR resources; for example, use dropdowns for ICD-10 codes, search for medications, and allow free-text notes. Validate fields on the client side; provide progress indicators when uploading data.
6. **Secure messaging** – Offer threaded conversations; allow attachments (lab reports, pictures); implement push notifications via Firebase Cloud Messaging or

Apple Push Notification Service; store messages in FHIR **Communication** resources.

7. **Notifications & reminders** – Provide configurable reminders for appointments and medication. For privacy, notifications should be generic (e.g., “You have a new message in the Health Card app”).

7 Back-end implementation details

1. **Microservices architecture** – Organise services by domain (e.g., user service, FHIR service proxy, appointment service, messaging service). Each service has its own database or uses FHIR as the source of truth. Use Docker and Kubernetes for container orchestration.
2. **API design** – Provide a RESTful API conforming to FHIR where possible. Additional endpoints can wrap FHIR operations (e.g., **POST** `/patients/{id}/encounters` creates an encounter). Implement GraphQL if flexible queries are needed. Use asynchronous messaging (e.g., RabbitMQ or Apache Kafka) for events like new encounter creation to trigger notifications.
3. **Database** – Use **PostgreSQL** with Transparent Data Encryption for non-FHIR tables (user accounts, audit logs). FHIR data resides in the HAPI FHIR database (which can be configured to use PostgreSQL or MySQL). Use **Elasticsearch** for full-text search across patient records.
4. **Caching & performance** – Use Redis for session storage and caching frequently accessed data (e.g., dictionary lists). Implement rate limiting at the API gateway.
5. **CNIC verification module** – Integrate NADRA API with strong encryption. The module sends CNIC to NADRA, optionally matches biometrics, and receives demographic data to populate the patient record. Caching results avoids repeated calls.
6. **Identity and access management** – Use an OpenID Connect provider (e.g., Keycloak or Auth0) to handle authentication, MFA and RBAC. Configure identity claims to include user role and organisation.
7. **Monitoring & logging** – Implement centralised logging (e.g., with ELK stack) and metrics collection (Prometheus + Grafana). Set up alerts for unusual access patterns and service failures.

8 Development roadmap and phases

1. **Discovery & planning (1–2 months)**

- Conduct stakeholder interviews (government, NADRA, hospitals, telemedicine providers, patient advocacy groups).
 - Define minimum viable product (MVP) scope and success metrics (e.g., number of patients enrolled, number of consultations recorded).
 - Draft detailed user stories and data flow diagrams.
- 2. Architecture design & prototyping (2 months)**
- Select technology stack (Flutter vs React Native; Node.js vs Spring Boot; database choices).
 - Deploy HAPI FHIR server in staging environment; set up CNIC verification sandbox.
 - Build wireframes and interactive prototypes for patient and doctor apps; conduct usability testing with sample users.
- 3. Core development (4–6 months)**
- **Back-end:** Implement identity service, FHIR proxy service, appointment service, messaging service and admin portal. Integrate NADRA API. Set up RBAC.
 - **Front-end:** Develop patient app features (registration, health timeline, QR scanning, messaging, appointments) and doctor app features (patient search, add notes/medications, view timeline). Implement multi-language support.
 - **Security & compliance:** Implement encryption, MFA, audit logging, data retention policies. Create standard consent forms and privacy policies.
- 4. Integration & interoperability (1–2 months)**
- Connect to pilot hospitals' EHR systems via FHIR; map FHIR resources to local codes (SNOMED, ICD). Test retrieving lab results and medication lists.
 - Integrate telehealth video module and ensure notifications align with privacy guidelines.
- 5. Testing & quality assurance (2 months)**
- Conduct unit, integration and performance tests. Perform security penetration testing and threat modelling. Validate compliance with PDPB and HIPAA guidelines.
 - Pilot with a small number of clinics; gather feedback; adjust UI/UX.
- 6. Pilot & deployment (3 months)**
- Onboard selected healthcare providers and issue physical/digital health cards to patients. Provide training and support.
 - Monitor usage, performance and user satisfaction; collect feedback; fix bugs.
 - Prepare for scale-up to other regions; plan for offline kiosks in rural areas for enrollment.

7. Ongoing maintenance & enhancements

- Implement continuous integration/continuous deployment (CI/CD) pipelines; release regular updates.
- Add additional features such as insurance integration, analytics dashboards, integration with wearable devices and AI-powered decision support.
- Conduct periodic security and compliance audits and update to future PDPB regulations.

9 Conclusion

A CNIC-based health card system that uses FHIR-first design and robust security practices can transform healthcare delivery in Pakistan. By providing separate patient and doctor apps, integrating with NADRA for identity verification, and complying with the PDPB and HIPAA guidelines, the platform ensures secure, continuous, and patient-centred care. The blueprint above outlines the functional requirements, technical architecture, data model, security measures and development roadmap necessary for a software agency to start building a scalable, secure and user-friendly solution. Adhering to these guidelines will align the project with Pakistan's **One Patient One ID** initiative and global interoperability standards, while ensuring sustainability and adoption across diverse populations.